

IEEE VR 2023 Workshops

Workshop: Multi-modal Affective and Social Behavior Analysis and Synthesis in Extended Reality (MASSXR)

ORGANIZERS:

Zerrin Yumak
Utrecht University, The Netherlands

Funda Durupinar
University of Massachusetts Boston, USA

Oya Celiktutan
King's College, London, UK

Pablo Cesar
CWI and TU Delft, The Netherlands

Aniket Bera
Purdue University, USA

Mar Gonzalez-Franco
Google Labs, USA

ABSTRACT:

The objective of this workshop on Multi-modal Affective and Social Behavior Analysis and Synthesis in Extended Reality (MASSXR) is to bring together researchers and practitioners working in social and affective computing with those in 3D computer vision and computer graphics/animation and discuss the current state and future directions, opportunities, and challenges. The workshop aims to establish a new platform for the development of immersive embodied intelligence at the intersection of Artificial Intelligence (AI) and Extended Reality (XR). We expect that the workshop will provide an opportunity for researchers to develop new techniques and lead to new collaborations among the participants.

ABOUT THE WORKSHOP:

The recent advances in immersive technologies such as realistic digital humans, off-the-shelf XR devices that can capture users' speech, face, hands, and body, together with the development of sophisticated data-driven AI algorithms are offering a great opportunity for the automatic analysis and synthesis of social and affective cues in XR, which are essential for truly user-aware interaction systems. Although affective and social signal understanding and synthesis are studied in other fields (e.g., for human-robot interaction, intelligent virtual agents, or computer vision), they have not yet been explored adequately in Virtual, Mixed, and Augmented Reality, demanding extended-reality-specific theoretical and methodological foundations. Particularly, this workshop focuses on the following research questions:

- How can we sense the user's affective and social states using sensors available in XR?
- How can we collect users' interaction data in immersive situations?
- How can we generate affective and social cues for digital humans/avatars in immersive interactions enabled by dialogue, voice, and non-verbal behaviors?
- How can we develop systematic methodologies and techniques to develop plausible, trustable, personalized behaviors for social and affective interaction in XR?

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graphics/animation and discuss the current state and future directions, opportunities, and challenges. The workshop aims to establish a new platform for the development of immersive embodied intelligence at the intersection of Artificial Intelligence (AI) and Extended Reality (XR). We expect that the workshop will enable cross-fertilization between different domains and provide an opportunity for researchers to develop new techniques and lead to new collaborations among the participants. The webpage of the workshop can be found at: <https://sites.google.com/view/massxrworkshop2023>.

SCOPE:

This workshop invited researchers to submit original, high-quality research papers related to multi-modal affective and social behavior analysis and synthesis in XR. Relevant topics include, but are not limited to:

- Analysis and synthesis of multi-modal social and affective cues in XR
- Data-driven expressive character animation (e.g., face, gaze, gestures, ...)
- AI algorithms for modeling social interactions with human- and AI-driven virtual humans
- Machine learning for dyadic and multi-party interactions
- Generating diverse, personalized, and style-based body motions
- Music-driven animation (e.g., dance, instrument playing)

- Multi-modal data collection and annotation in and for XR (e.g., using VR/AR headsets, microphones, motion capture devices, and 4D scanners)
- Efficient and novel machine learning methods (e.g., transfer learning, self-supervised and few-shot learning, generative and graph models)
- Subjective and objective analysis of data-driven algorithms for XR
- Applications in healthcare, education, and entertainment (e.g., sign language)

KEYNOTE SPEAKERS:

MASSXR features four keynote speakers with complementary research lines. Taku Komura has expertise on synthesis of animation of virtual humans. His talk is entitled “Building Virtual Humans for Real-Time Communication and Interaction”. Srikanth Narayanan has a research line on affective and social signal analysis. In this workshop, he will give a talk on “Multimodal Machine Intelligence: From Signals to Human-centered Experiences”. Sylvia Xueni Pan focuses on the human-centric aspects of embodiment in XR, particularly in the application area of health. Her talk is entitled “Immersive Social Interaction for Health and Healthcare”. Last but not least, Julie Williamson will discuss the role of social signals, such as position, posture, gesture, facial expression in her talk “Being Social in XR”. In summary, the keynote speakers are:

- Taku Komura, The University of Hong Kong, HK
- Sylvia Xueni Pan, Goldsmiths, University of London, UK
- Srikanth Narayanan, University of Southern California, USA
- Julie Williamson, University of Glasgow, UK

WORKSHOP CONTRIBUTIONS:

The workshop accepted one research paper and one work-in-progress paper.

- *“Augmented Reality and Affective Computing for Nonverbal Interaction Support of the Visually Impaired”* by Deniz Iren (Open Universiteit, NL) et al: Nonverbal cues such as gestures and facial expressions are indispensable in human communication. However, such an essential aspect of social interactions is inaccessible to the sight impaired. This issue can be alleviated with the assistance of augmented reality and affective computing techniques embodied as wearable technology. Even though both augmented reality and affective computing have been studied comprehensively, the real-life deployment and utilization of these techniques are hindered by the limitations of wearable devices in terms of computational capabilities and battery life. This calls for a holistic approach to implementing lightweight and robust affective computing methods. In this study, the authors present a prototype that combines facial expression and gesture recognition that is optimized to function on battery-powered wearable devices. Additionally, the prototype embodies a haptic sleeve that communicates the detected facial expressions and gestures to the wearer.
- *“Towards a multimodal VR trainer of voice emission and public speaking - work in progress”* by Magdalena Igras-

Cybulska (AGH UST, Kraków, Poland) et. al: GlossoVR is a virtual reality (VR) application that combines training in public speaking in front of a virtual audience and in voice emission in relaxation exercises. It is accompanied by digital signal processing (DSP) and artificial intelligence (AI) modules which provide automatic feedback on the vocal performance as well as the behavior and psychophysiology of the user. In particular, the authors address parameters of speech emotions, prosody and timbre, and the user's hand gestures and eye movement. This article reports the work in progress, focusing on the approaches, datasets and algorithms applied in the current state of the GlossoVR project.

PANEL:

MASSXR workshop has a panel discussion including the keynote speakers, organizers, and audience to discuss the current state, future directions, opportunities, and challenges in this area in an interactive manner.

INTERNATIONAL PROGRAM COMMITTEE:

The technical program committee was composed of 19 experts in the areas relevant to workshop theme.

- Abdallah El Ali, University of Amsterdam, The Netherlands
- Uttaran Bhattacharya, Adobe Inc
- Mark Billingham, University of South Australia, Australia
- Claudia Esteves, Universidad de Guanajuato, México
- Hayley Hung, TU Delft, The Netherlands
- Ugur Gudukbay, Bilkent University, Turkey
- Sameer Kishore, Middlesex University Dubai
- Stefan Kopp, Bielefeld University, Germany
- Dinesh Manocha, University of Maryland, USA
- Stacy Marsella, Northeastern University, USA
- Rachel McDonnell, Trinity College Dublin, Ireland
- Aline Normoyle, Bryn Mawr College, USA
- Sylvie Xueni Pan, Goldsmiths, University of London, UK
- Catherine Pelachaud, Sorbonne University, France
- Voicu Popescu, Purdue University, USA
- Albert Ali Salah, Utrecht University, The Netherlands
- Ana Tajadura-Jiménez, University College London, UK
- Nguyen Tan Viet Tuyen, King's College London, UK
- Julie Williamson, University of Glasgow, UK